

## A Variational Method for Discovering Multiple Sparse Solutions in Classification and Regression Problems

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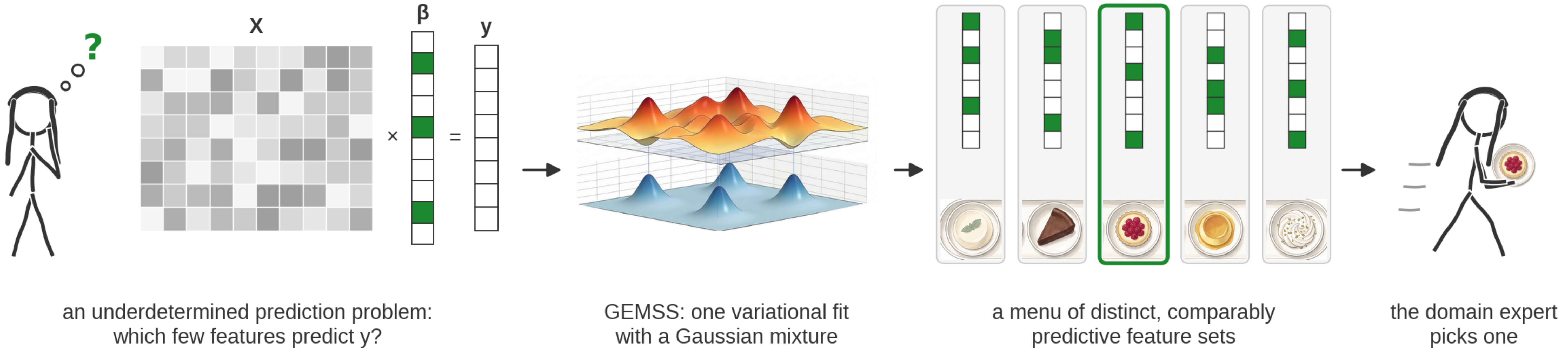
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Use cases: Expensive sample measurements • Domain expertise matters • Correlated or too many features



### GEMSS

= Gaussian Ensemble for Multiple Sparse Solutions

**The starting point:** underdetermined problem, generalized linear model

**The algorithm:**

**1 Prior:** Structured Spike & Slab, exactly  $D$  nonzeros.

**2 Approximation:** a mixture of  $K$  Gaussians, one component per mode:

$$q(\beta) = \sum_{k=1}^K \alpha_k \mathcal{N}(\beta; \mu_k, \text{diag } \sigma_k^2)$$

**3 One objective, one fit:** maximize ELBO over mixture parameters, by SGD:

$$\mathcal{L} = \mathbb{E}_q[\log p(y | X, \beta)] - \text{KL}(q || \pi)$$

**4 Read-out:** one component  $\Rightarrow$  one candidate solution: largest  $|\mu_k|$  values.

### A novel benchmark

Generate problems with multiple known solutions, evaluate recovery

**Why:** prediction metrics alone cannot tell if practically usable solutions were found.

**Generator:**

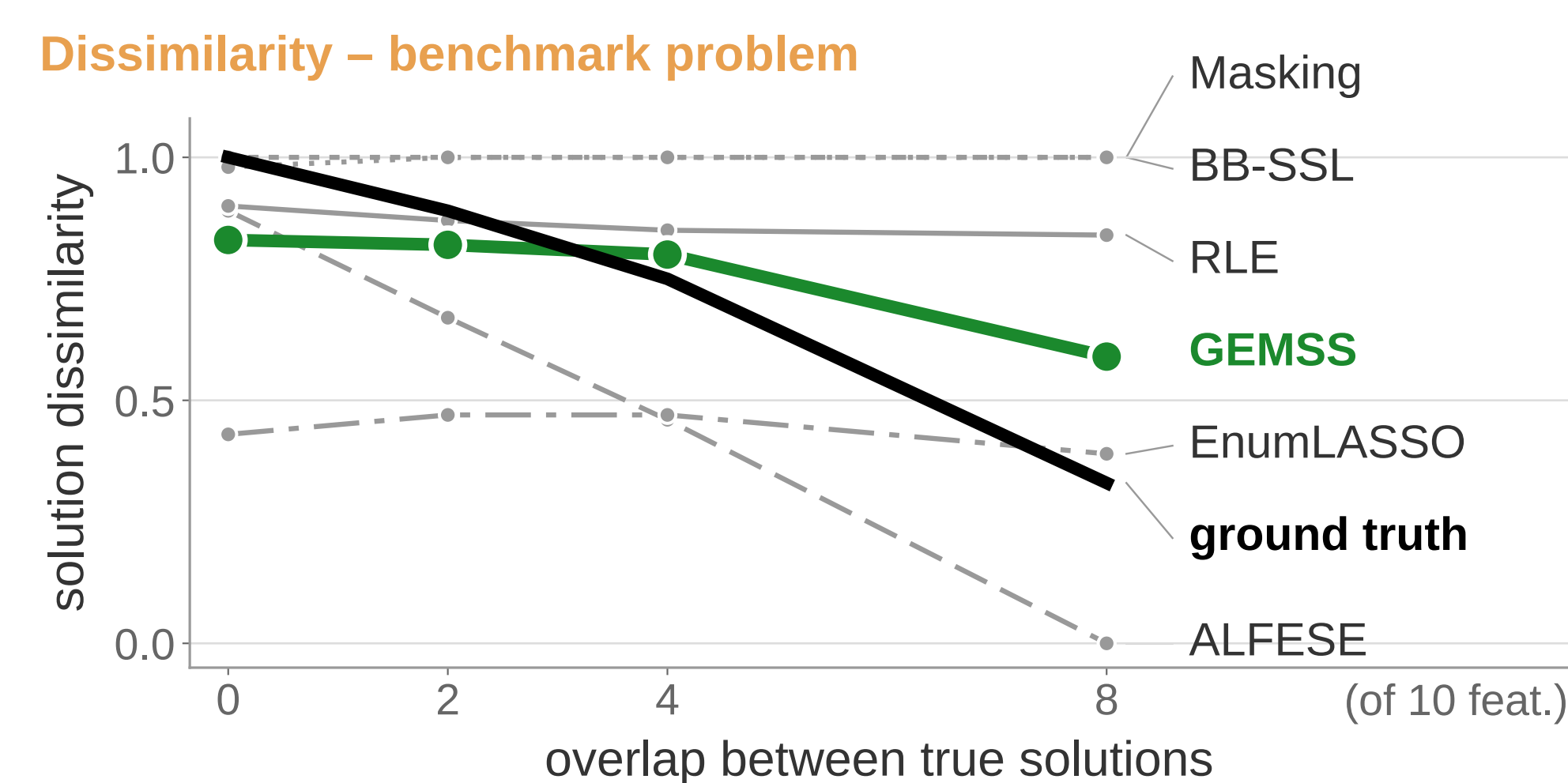
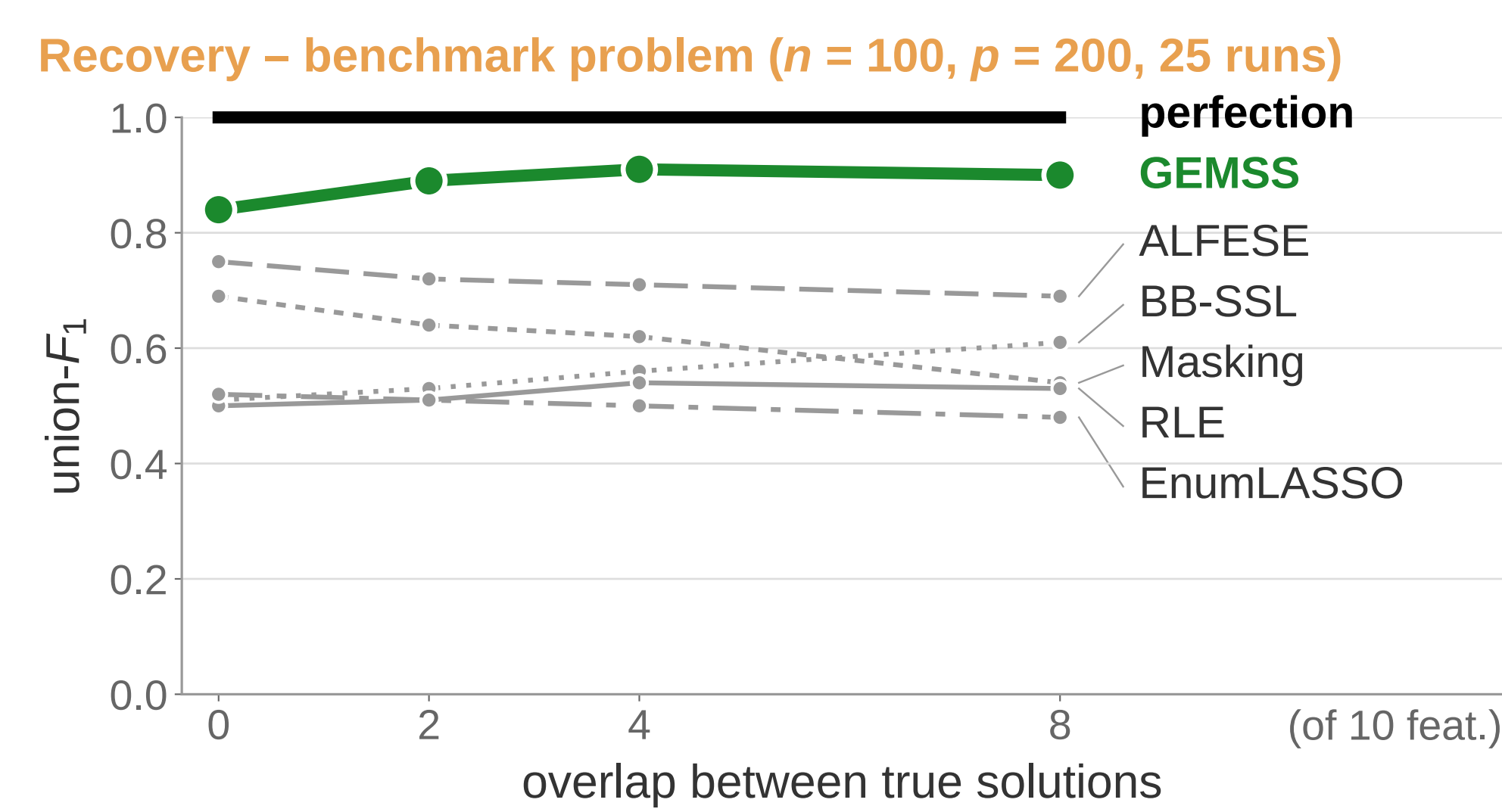
- Rank- $r$  latent mechanism plants several solutions that are equally predictive by construction ( $R^2 \sim 1$ ).
- Parameter *overlap* defines solutions' shared core of features.
- Optional: noise, missingness, class imbalance.
- Binimize  $y$  for classification.

**Evaluation metrics:**

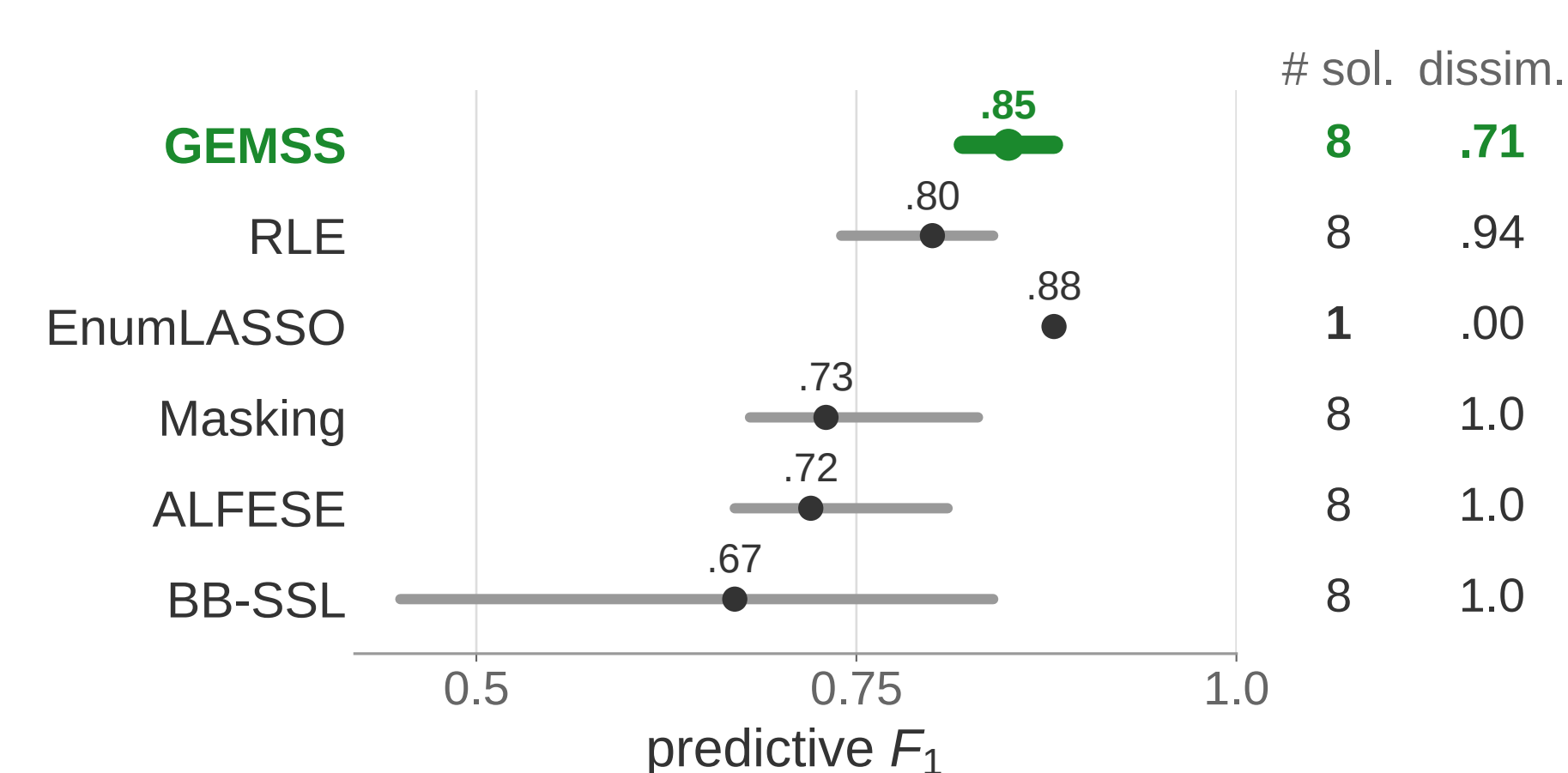
- Union- $F_1$ :** are all planted features recovered?  $\rightarrow$  compare pooled features
- Dissimilarity:** how distinct are the solutions?  $\rightarrow$  read against ground truth

## Experimental results

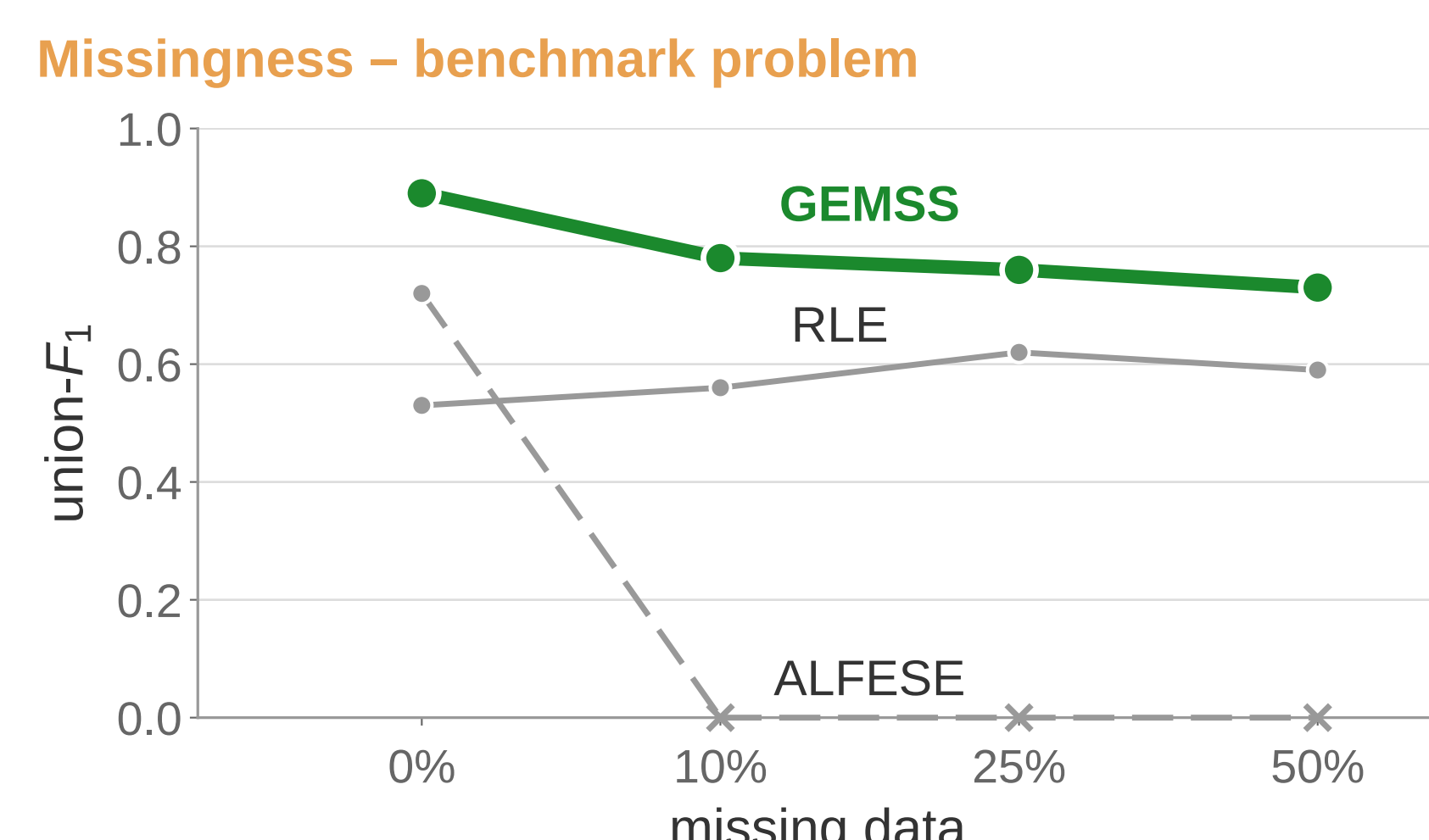
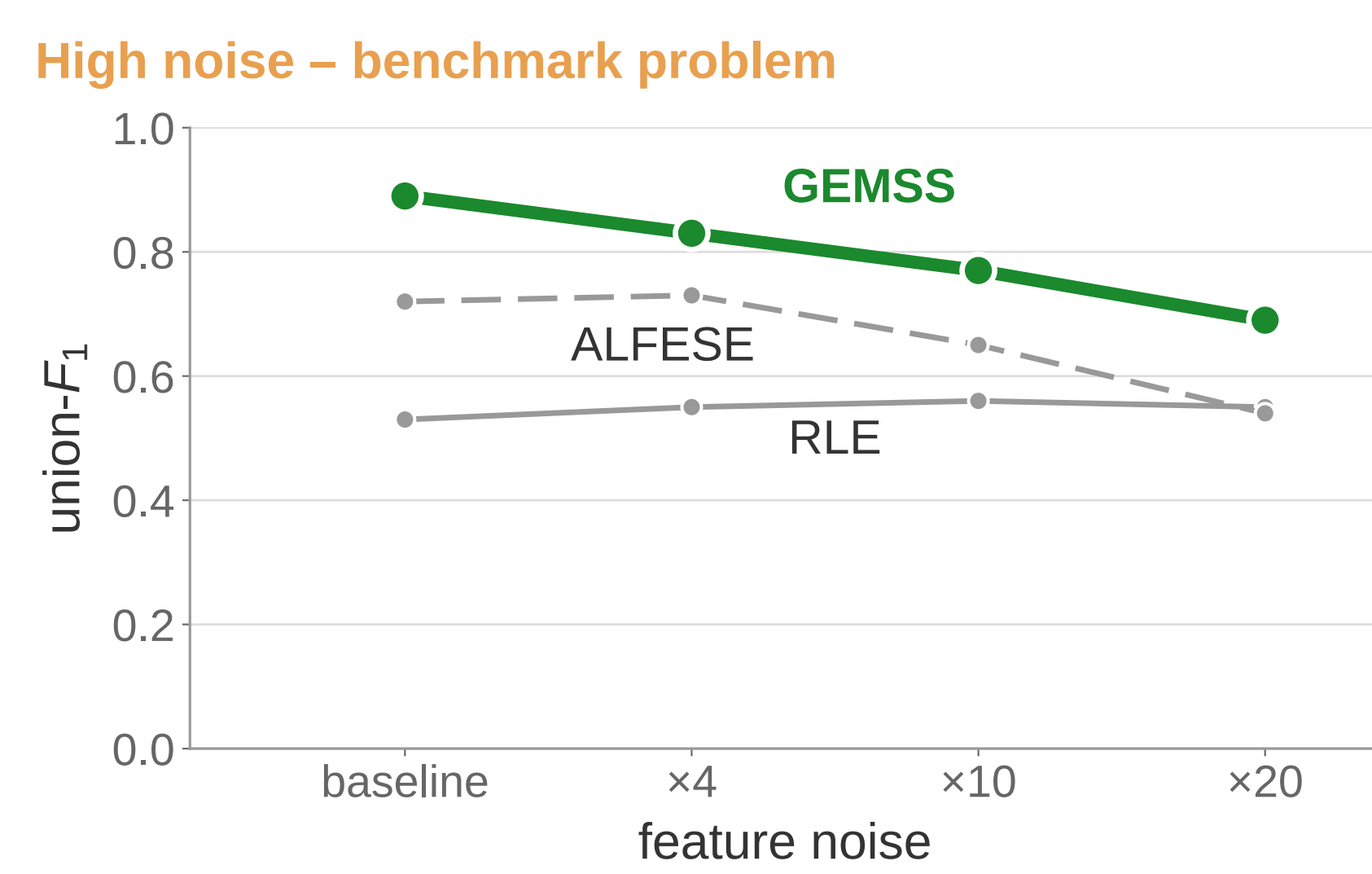
### GEMSS outperforms its competitors, on synthetic and on real data.



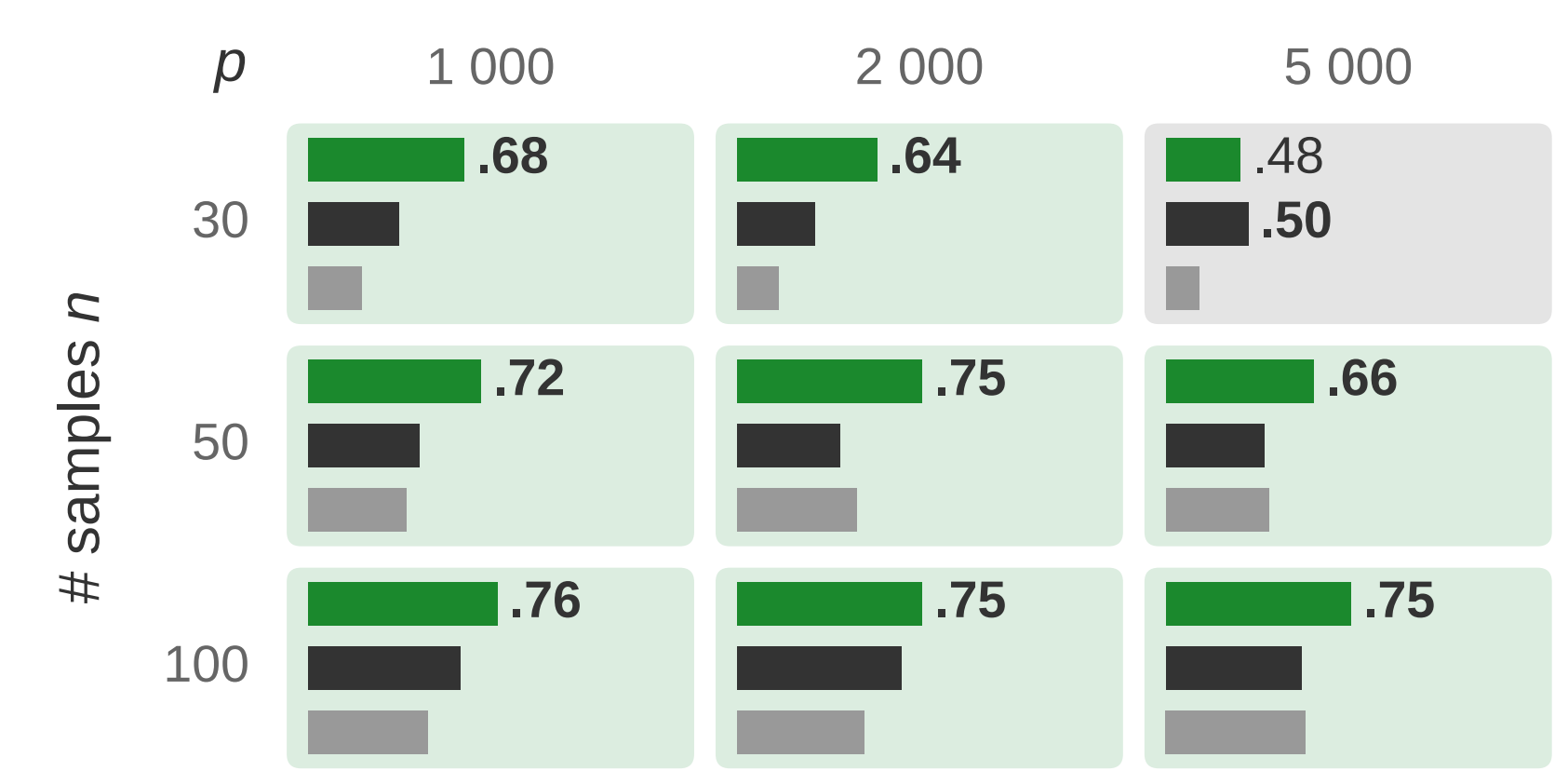
**Preterm birth biomarkers ( $n = 149, p = 487, \text{requested } K = 8 \text{ sol.}$ )**



### GEMSS leads under stress: noise, missing data, dimension.

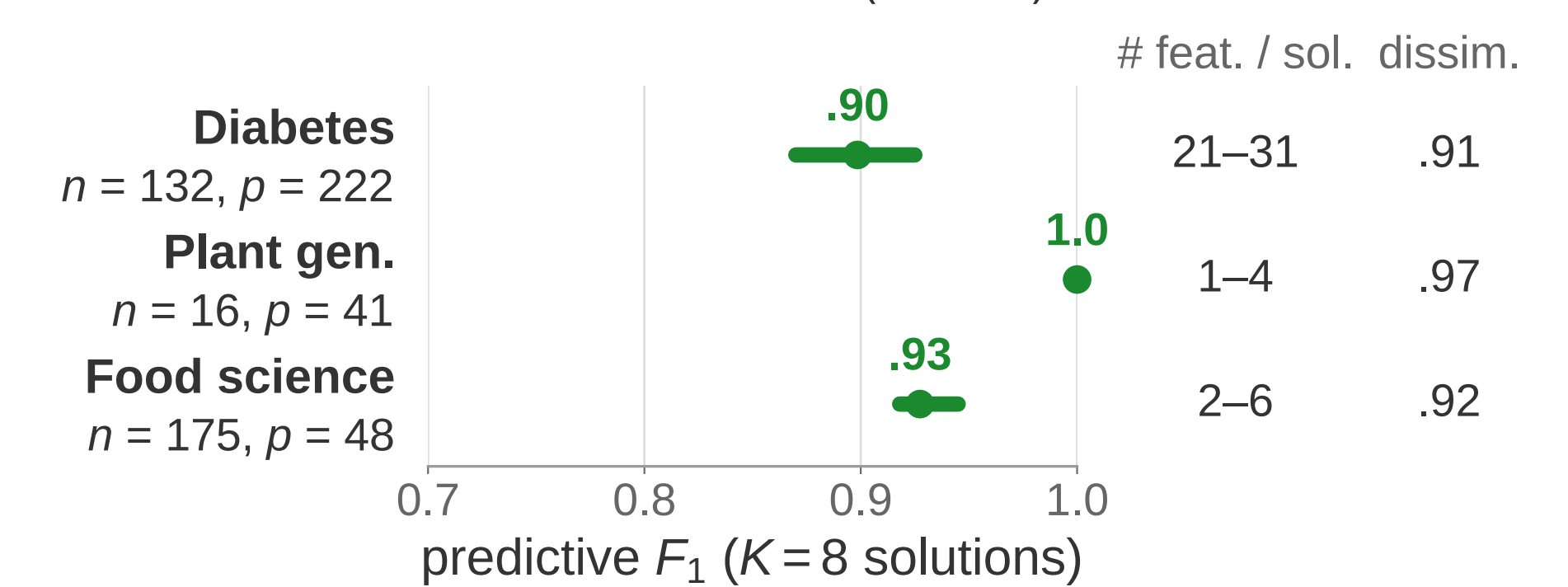


**Dimensionality sweep, overlap = 6/10:**  
union- $F_1$  of GEMSS (green), ALFESE (dark), RLE (light)



### Verified: GEMSS is applicable to challenging real-world data.

- Diabetes biomarkers:** primary use case  $n < p$ , urinary metabolites as biomarkers.
- Plant genomics:** extremely low sample size  $n = 16$ , metabolomics of *Arabidopsis thaliana*.
- Food science:** case  $n > p$  with highly correlated features, class imbalance (11/89).



Solutions checked against publications (MetaboLights) or by a domain expert (food science).

**Methods:** RLE = Randomized-LASSO Ensemble (Meinshausen & Bühlmann, JRSS-B 2010) · ALFESE = Alternative Feature Selection (Bach & Böhm, IJSDA 2024) · BB-SSL = Bayesian-Bootstrap Spike-and-Slab LASSO (Nie & Ročková, JASA 2023) · EnumLASSO = Enumeration LASSO (Hara & Maehara, AAAI 2017) · Masking = Naive iterative masking

**Data sources:** Diabetes, Plant genomics, Preterm birth: MetaboLights archive · Food science: proprietary

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